CODING SHUTTLE

Docker

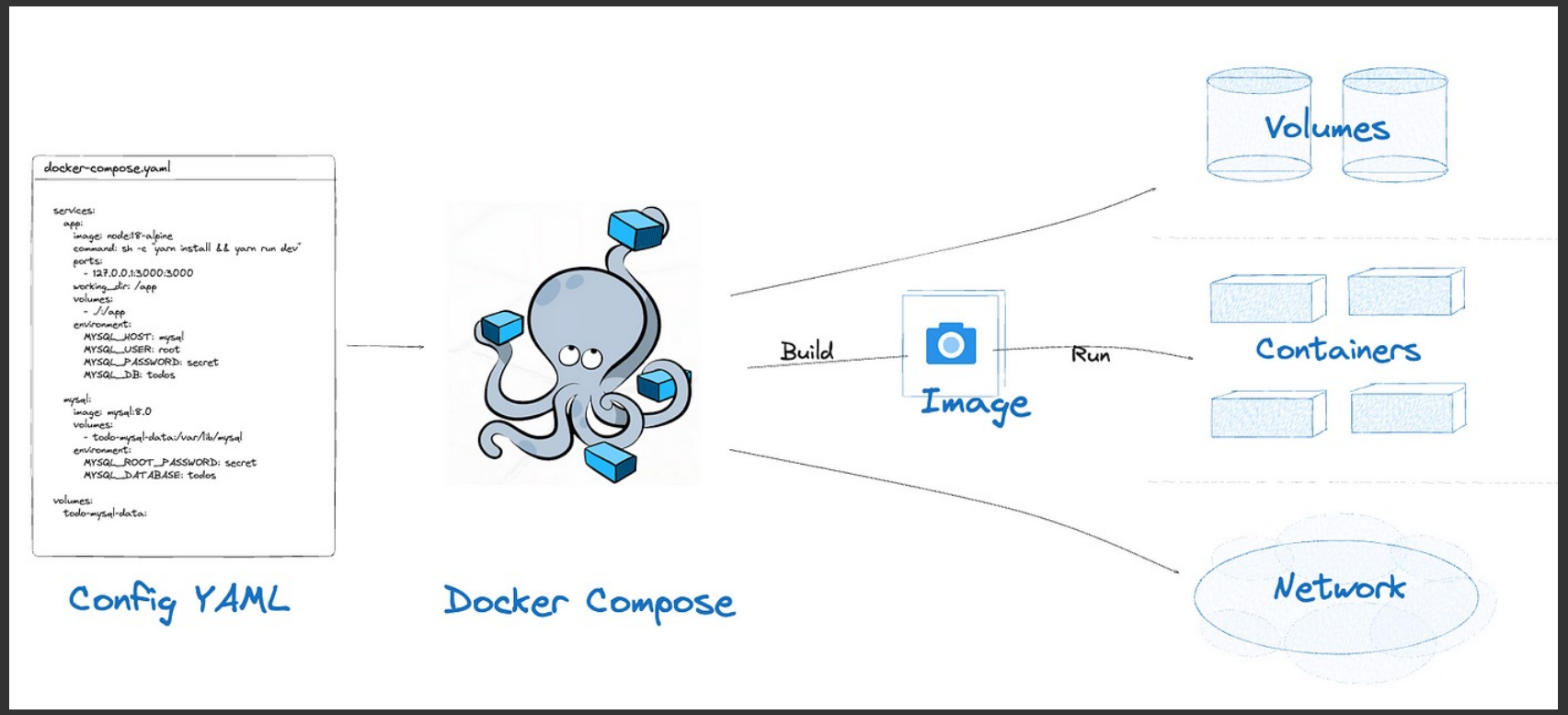
Docker Compose

Docker Compose

Docker Compose is a tool that makes it easier to create and run multi-container applications. It automates the process of managing several Docker containers simultaneously, such as a website frontend, API, and database service.

Docker Compose allows you to define your application’s containers as code inside a YAML file you can commit to your source repository. Once you’ve created your file (normally named docker-compose.yml), you can start all your containers (called “services”) with a single Compose command.

Using the Dockerfile



Docker vs Docker-compose

Docker is a containerization engine that provides a CLI for building, running, and managing individual containers on your host.

Compose is a tool that expands Docker with support for multi-container management. It supports “stacks” of containers that are declaratively defined in project-level config files.

Creating docker-compose.yml

Define the following:

Version

Services

Networks

volumes

Example of docker-compose.yml

```yaml
version: '3.8'
services:
  web:
    image: nginx:latest
    ports:
      - "8080:80"
  db:
    image: postgres:latest
    environment:
      - POSTGRES_DB=mydb
      - POSTGRES_USER=user
      - POSTGRES_PASSWORD=password
    volumes:
      - db-data:/var/lib/postgresql/data
    ports:
      - "5432:5432"
volumes:
  db-data:
```

Docker compose yml services

services:

- image:
- container_name:
- ports: Maps host machine ports to container ports (e.g., "8080:80" maps port 8080 on the host to port 80 in the container).

- volumes: Mounts directories or files from the host into the container
- environment: Set environment variables inside the container (e.g., POSTGRES_DB=mydatabase).

- depends_on: Ensures that one service starts before this one
- networks: Specifies which networks the service will connect to.

Docker-compose commands

```
docker-compose up

docker-compose down

docker-compose start

docker-compose stop

docker-compose build

docker-compose ps

docker-compose logs <service-name>
```